

# International Transfers or Differentiated Carbon Prices?

Adrien Fabre\*

Constance Gorge<sup>†‡</sup>

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## Abstract

Recent proposals for international carbon pricing differentiate national carbon prices by income. The usual justification is that cross-border transfers are politically infeasible. But differentiated prices are themselves equivalent to transfers. We show that any schedule of national prices is distributionally equivalent to a uniform price combined with a certain allocation of tradable emission rights, equal at the first order to each country's emissions under price differentiation. In a dynamic setting, the equivalent rights are weighted by the discounted carbon price, and under Hotelling pricing they reduce to a carbon budget. Because abatement costs are convex, this allocation leaves every country weakly better off, so the uniform price with equivalent rights Pareto-dominates differentiation. We quantify the equivalence in NICE2020, an integrated assessment model with 179 countries. Replacing the schedules of [Parry et al. \(2021\)](#), [Wolfram et al. \(2025\)](#) or [Banerjee et al. \(2025\)](#) with a uniform price and equivalent rights would cut the coalition's emissions by about 5% while leaving every member as well off. Alternatively, holding emissions fixed, it would raise every member's consumption. We also review the departures from standard assumptions that could justify price differentiation and argue that they are not compelling.

**Keywords:** carbon pricing; international climate agreements; emissions trading; burden sharing; differentiated carbon prices; international transfers; integrated assessment.

**JEL:** Q54; Q58; H23; F53; D61.

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\*CNRS; CIRED. Corresponding author: [adrien.fabre@cnrs.fr](mailto:adrien.fabre@cnrs.fr).

<sup>†</sup>CIRED.

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# Contents

|          |                                                                          |           |
|----------|--------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                      | <b>2</b>  |
| <b>2</b> | <b>Why differentiate carbon prices?</b>                                  | <b>4</b>  |
| <b>3</b> | <b>The equivalence between differentiated prices and emission rights</b> | <b>6</b>  |
| 3.1      | Static setting . . . . .                                                 | 6         |
| 3.2      | Dynamic setting . . . . .                                                | 8         |
| <b>4</b> | <b>Equivalent rights by country</b>                                      | <b>10</b> |
| 4.1      | Model . . . . .                                                          | 10        |
| 4.2      | Design . . . . .                                                         | 11        |
| 4.3      | Results . . . . .                                                        | 11        |
| <b>5</b> | <b>Equivalent rights of prominent differentiation proposals</b>          | <b>14</b> |
| 5.1      | Proposals . . . . .                                                      | 14        |
| 5.2      | Design . . . . .                                                         | 15        |
| 5.3      | Results . . . . .                                                        | 16        |
| <b>6</b> | <b>Conclusion</b>                                                        | <b>19</b> |
| <b>A</b> | <b>Model details</b>                                                     | <b>23</b> |
| <b>B</b> | <b>Solves, sharing rules and numerical accuracy</b>                      | <b>24</b> |
| <b>C</b> | <b>Welfare or consumption, and the Equal Right schedule</b>              | <b>29</b> |

## 1 Introduction

International climate policy is stuck on a distributional question: who should obtain the rights to emit? The Paris Agreement endorses common but differentiated responsibilities and respective capabilities, and it commits developed countries to financing climate action in developing ones. It does not translate these principles into numbers, and national contributions remain far short of its temperature goal (UNEP 2025).

A prominent recent answer is to coordinate carbon prices while differentiating them by income. The IMF proposes an international price floor of \$75, \$50 and \$25/tCO<sub>2</sub> for high-, upper middle- and lower-income countries (Parry et al. 2021). Wolfram et al. (2025) propose a climate coalition built on the same tiers. Banerjee et al. (2025) propose a “grand bargain” in which low-, and middle-income countries adopt prices of \$10 to \$50/t in exchange for the compensation

of their climate damages. In all three proposals, carbon revenues are retained domestically. Differentiation is meant to deliver equity without the cross-border transfers that are deemed politically infeasible (Parry et al. 2021).

Textbook theory prescribes something else: a uniform price for efficiency, and lump-sum transfers for equity (Aldy & Stavins 2012). Differentiated prices are a second-best substitute for transfers, and the case for income-based differentiation therefore rests on the premise that transfers are infeasible. This paper argues that the premise does not rest on solid ground. Survey evidence already challenges it: majorities accept an international carbon price financing North–South transfers in all 23 surveyed countries (Fabre 2025; Fabre et al. 2025). Furthermore, a uniform price can achieve efficiency while remaining distributionally equivalent to differentiated prices. We make the claim precise in three results.

First, any schedule of differentiated autarky prices is distributionally equivalent to a uniform price combined with a specific allocation of tradable emission rights. To first order, this allocation grants each country the emissions it has under price differentiation. A country allowed a price below the uniform price therefore receives more rights than it emits at the uniform price, and sells the difference. Conversely, a positive price gap is equivalent to less emission rights than in autarky. In a static setting, we derive a second-order formula in terms of the price gap and the price elasticity of emissions (Proposition 1).

Second, granting each country its emissions under the schedule leaves every country weakly better off under the uniform price than under differentiation (Proposition 2). This is the familiar gains-from-trade logic of emissions markets (Montgomery 1972) and the international analogue of production efficiency (Diamond & Mirrlees 1971). Its implication for the policy debate is less familiar: any price-differentiation agreement can be replaced by a uniform price with an allocation of rights such that no country loses. The resulting surplus can take the form of reduced emissions or of increased consumption.

Third, in a dynamic setting the equivalence holds in present value, with each year’s rights weighted by the discounted carbon price (Proposition 3). When the carbon price follows a Hotelling path, the equivalent allocation is a carbon budget: each country’s cumulative emissions under price differentiation.

We quantify the equivalence in NICE2020 (Young-Brun et al. 2025), an integrated assessment model with 179 countries, in two exercises. First, for eight diverse economies, we compare welfare under a differentiated autarky price and under a global uniform price with a given allocation, holding global emissions on a path that exhausts a carbon budget of 1000 GtCO<sub>2</sub> over 2025–2100. For each autarky price and each allocation, we compute the welfare difference and trace the indifference curve between the two regimes. The curves illustrate that the equivalence runs only one way: many allocations of rights, including an equal per capita allocation, cannot be replicated by any schedule of non-negative prices, because a zero price is not enough to compensate low emitters. For example,

Nigeria would prefer a uniform price to any non-negative autarky price, including a zero price, as long as its emission rights per capita are at least 40% of the global average.

Second, we compute the equivalent rights for prominent differentiated prices proposals. Granting all members their equivalent rights cuts coalition emissions by 4.7% (Banerjee et al. 2025), 5.1% (Wolfram et al. 2025) and 15.2% (Equal Right 2023), while leaving every member as well off as under the schedule.

The remainder of the paper is organised as follows. Section 2 reviews the rationales for differentiated carbon prices in a world that departs from standard assumptions. Section 3 derives the equivalence between differentiated prices and emission rights, in closed form in the static case and under additional assumptions in the dynamic case. Section 4 uses NICE2020 to trace, for eight major economies, the rights that leave each one indifferent to any given autarky price. Section 5 computes the rights equivalent to prominent differentiated-price proposals, and the emissions cut or consumption gain obtained by replacing them with a uniform price. Section 6 concludes.

## 2 Why differentiate carbon prices?

With complete markets and no other distortion, efficiency requires a uniform carbon price, and equity should be addressed via lump-sum transfers (Sandmo 2007/oct./01). Four departures from standard assumptions have been invoked to justify differentiating carbon prices across countries or sectors: divergent discount rates, pre-existing distortive taxes, market power in trade, and constraints on cross-country transfers. We review them in turn.

First, Anthoff et al. (2026) show that equalising carbon prices across countries is inefficient in case of ‘cross-country differences in discount rates (or interest rates)’, which arise from inefficiencies in the allocation of capital. By Hotelling’s argument, each country’s carbon price should grow at its own discount rate, so prices cannot be equal at all dates.<sup>1</sup> The authors stress, however, that their result cannot say whether trade in emission permits from a given allocation would raise or lower welfare, as this depends on how permit trade interacts with the capital-market frictions, which no model has yet specified.

Second, pre-existing tax distortions can be mitigated by differentiating carbon prices across sectors or countries (Babiker et al. 2004; Boeters 2014). For example, if aviation remains under-taxed, a higher carbon price on aviation than on other sectors improves welfare. For this reason (applied to countries), Babiker et al. (2004) show that a uniform price may not always be beneficial to the countries

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<sup>1</sup>Differentiated prices may then be optimal in a second-best world, but the first best is in principle attainable by correcting the capital-market imperfections themselves.

that take part in it. However, [D'autume et al. \(2016\)](#) show that a uniform price remains optimal if the other tax (such as a gasoline tax) addresses a local externality or if public goods have to be funded by distortive taxation.

Third, large countries have market power in world markets, which gives them a terms-of-trade motive to differentiate ([Babiker 2005](#); [Boeters 2014](#)). Take China, a large oil importer and manufacturing exporter. By taxing oil more heavily than other goods, China would depress world oil demand and hence the oil price, improving its terms of trade. Similarly, it has an interest in setting a higher carbon price on manufactured goods, to raise the price of its exports (to the extent the carbon price paid by importers exceeds the loss of exports due to higher prices).

Fourth, and most commonly, differentiated prices are justified by the assertion that international transfers are politically infeasible ([Parry et al. 2021](#); [Young-Brun et al. 2025](#)). [Bauer et al. \(2020\)](#) describe an “efficiency–sovereignty” trade-off in climate policy: either carbon prices are differentiated and efficiency is lost, or a uniform price with transfers is applied, hampering national sovereignty. This reasoning assumes that international transfers constitute a stronger infringement on sovereignty than coordinated price differentiation (see also [Bekkers & Caricola 2022](#)). Because transfers are deemed infeasible for they infringe sovereignty, differentiated prices are viewed as a necessary second-best solution.

The first three rationales have an efficiency basis and do not rest on the absence of transfers. They also call for a differentiation that bears no systematic relation to income: by discount rate, by sector or by market power. This paper addresses the fourth rationale, the one invoked by differentiation proposals. We show that it cannot rest on the distributional implications of transfers, since differentiation entails implicit costs or benefits of the same size as the transfers that a uniform price with equivalent rights would involve.

Our analysis builds on the literature that evaluates specific differentiated price floors. In a global general equilibrium model, [Chateau et al. \(2024\)](#) find that the IMF’s international carbon price floor ([Parry et al. 2021](#)) achieves higher welfare than a uniform price *without transfers* attaining the same emission reductions (at \$56/t), since it shifts part of the mitigation effort from low- to high-income countries. They also find that it protects the competitiveness of high-income countries as well as a border carbon adjustment would. [He et al. \(2024\)](#) show that, relative to national pledges, the IMF price floors bind on China and India but not on high-income countries, which makes the arrangement unattractive to developing countries.

[Le Moigne et al. \(2026\)](#) find that income-differentiated floors do not necessarily yield fairer outcomes. Compared with a uniform tax accompanied by the North–South transfers that equalise costs across countries (\$107–169 billion a year), all variants of the IMF floor entail larger global real-income losses, and most of them require larger transfers to reach the same fairness. We provide the first mapping from such schedules to equivalent allocations of emission rights.

Our results also speak to the literature on negotiating a common carbon price or a global cap-and-trade with revenue sharing (Cramton et al. 2017; Fabre 2024; Gollier & Tirole 2015; Grubb 1990; Nordhaus 2015; Weitzman 2014). The equivalence formula gives climate negotiators a closed-form exchange rate between price concessions and emission rights.

### 3 The equivalence between differentiated prices and emission rights

#### 3.1 Static setting

Consider countries  $i = 1, \dots, I$  with populations  $N_i$ , in a single period. Lower-case variables are per capita. Country  $i$  emits  $e_i$  at abatement cost  $a_i(e_i)$ , which is decreasing and convex in  $e_i$ . Facing a carbon price  $p$ , firms and households emit  $e_i(p)$ , optimising by equating the marginal abatement cost to (minus) the price:  $a'_i(e_i(p)) = -p$ . Consumption is

$$c_i = y_i - a_i(e_i) + t_i, \quad (1)$$

where  $y_i$  is potential output and  $t_i$  a net international transfer. Welfare is increasing in  $c_i$  and decreasing in global emissions through climate damages.

We compare two regimes that deliver the same global emissions  $E = \sum_j N_j e_j$ , whose world average per capita is  $\bar{e} \equiv E/N$  (where  $N$  is the world population). Damages are then identical across regimes, and a country is better off in one regime if and only if its consumption is higher.

**Differentiated prices (autarky).** Country  $i$  applies a domestic price  $p_i$ , keeps its revenue and receives no transfer. It emits  $e_i^A \equiv e_i(p_i)$ , and  $\sum_i N_i e_i^A = E$ .

**Uniform price.** The same countries apply a uniform price  $p^*$  and trade emission rights under a cap equal to  $E$ . Country  $i$  receives rights  $r_i$  per capita. At the market-clearing price,  $\sum_i N_i e_i(p^*) = E$ , and  $i$  emits  $e_i^* \equiv e_i(p^*)$ . Its net transfer is the value of its sales minus its purchases of rights,  $t_i = (r_i - e_i^*) p^*$ .

Throughout, a star denotes the uniform-price regime, a superscript  $A$  the differentiated regime and a bar a world average. We write  $\pi_i \equiv p_i/p^*$  for country  $i$ 's relative price, and  $\varepsilon_i \equiv -\frac{p^*}{e_i^*} e'_i(p^*)$  for the price elasticity of its emissions at the uniform price. Define

$$b_i \equiv a_i(e_i^A) - a_i(e_i^*) - a'_i(e_i^*) (e_i^A - e_i^*) \geq 0. \quad (2)$$

$b_i$  is the deadweight loss of the price wedge  $p_i - p^*$ , i.e. the Harberger triangle of country  $i$ : the area enclosed by the country's marginal abatement cost curve, the horizontal line at  $p^*$  and the vertical line at  $e_i^A$ . Formally,  $b_i$  is the Bregman divergence of  $a_i$  between  $e_i^A$  and  $e_i^*$ . It is nonnegative by convexity, and strictly positive if  $a_i$  is strictly convex and  $p_i \neq p^*$ .

Denote  $r_i^*$  the *equivalent rights* that leave a country  $i$  exactly as well off under the uniform-price regime as under the differentiated regime with a price  $p_i$ .

**Proposition 1** (Equivalent rights). *Holding total emissions at  $E$  and the carbon price at  $p^*$ , we have*

$$r_i^* = e_i^A - \frac{b_i}{p^*}. \quad (3)$$

To first order in the price gap,

$$r_i^* \approx e_i^A \approx e_i^* \left[ 1 + \varepsilon_i(1 - \pi_i) \right]. \quad (4)$$

To second order, with  $\varepsilon_i$  evaluated at  $p^*$ ,

$$r_i^* \approx e_i^A - \frac{(e_i^A - e_i^*)^2}{2 \varepsilon_i e_i^*} \approx e_i^* \left[ 1 + \varepsilon_i(1 - \pi_i) - \frac{1}{2} \varepsilon_i(1 - \pi_i)^2 \right]. \quad (5)$$

*Proof.* Equating  $y_i - a_i(e_i^*) + (r_i - e_i^*)p^*$  with  $y_i - a_i(e_i^A)$  gives  $r_i = e_i^* + [a_i(e_i^*) - a_i(e_i^A)]/p^*$ . Substituting  $a_i'(e_i^*) = -p^*$  into (2) yields (3). Equation (5) follows from  $e_i^A - e_i^* \approx \varepsilon_i e_i^*(1 - \pi_i)$ . For (5), note that  $a_i''(e_i^*) = -1/e_i'(p^*) = p^*/(\varepsilon_i e_i^*)$ , so that  $b_i \approx \frac{1}{2} p^* (e_i^A - e_i^*)^2 / (\varepsilon_i e_i^*)$ .  $\square$

Proposition 1 shows that a differentiated price is a transfer paid in kind. Relative to the uniform price, a country allowed a lower price ( $\pi_i < 1$ ) keeps emissions  $e_i^A > e_i^*$ . To first order, the uniform-price regime replicates its situation by granting it exactly those emissions as rights, part of which it then sells, receiving  $(e_i^A - e_i^*)p^*$ . Conversely, a country asked for a higher price ( $\pi_i > 1$ ) is in the position of a country that buys rights. The size of the implicit transfer grows with the price gap, the country's emissions and the elasticity of those emissions. The same formula, read in reverse, converts any allocation of rights into an equivalent autarky price.

It is convenient to express an allocation as a multiple of the world average,  $r_i = \rho_i \bar{e}$ , so that  $\rho_i = 1$  is an equal per capita allocation. We denote by  $\rho_i^* \equiv r_i^*/\bar{e}$  the equivalent ratio, and by  $\hat{\rho}_i$  its first-order approximation:

$$\hat{\rho}_i \equiv \frac{e_i^A}{\bar{e}}, \quad (6)$$

country  $i$ 's per capita emissions under the differentiated schedule relative to the world average. By (3),  $\rho_i^* = \hat{\rho}_i - b_i / (p^* \bar{e}) \leq \hat{\rho}_i$ : the first-order prediction errs on the generous side. A differentiated schedule is thus, to first order, a grandfathering rule, applied to the emissions that the schedule itself induces.

**Proposition 2** (Dominance of the uniform price). *Allocate rights  $r_i = e_i^A$  to every country. This allocation (i) is feasible at the cap of the differentiated regime,  $\sum_i N_i r_i = E$ ; (ii) leaves every country weakly better off than under the differentiated regime, strictly so if  $a_i$  is strictly convex and  $p_i \neq p^*$ ; and (iii) leaves a surplus of  $\sum_i N_i b_i \geq 0$ . Expressed in emissions, the rights can be cut to the equivalent allocation  $(r_i^*)_i$ , with no country worse off than under differentiated prices.*

*Proof.* Part (i) holds by construction. For part (ii), the consumption gain of country  $i$  is  $[a_i(e_i^A) - a_i(e_i^*)] + (e_i^A - e_i^*)p^* = b_i \geq 0$ . For part (iii), sum these gains.  $\square$

A country that is allowed a lower price saves on abatement costs. Under the uniform price, the same country can abate up to  $p^*$  and sell the freed rights at  $p^*$ , which is always at least as good a deal. By Proposition 1, the surplus is of second order in the price gaps, whereas the implicit transfers are of first order. The efficiency argument for a uniform price is therefore robust but quantitatively modest when prices are not very dispersed, whereas the distributional stakes of the allocation are large.

Proposition 2 holds total emissions at  $E = \sum_i N_i e_i^A$  and the uniform price at  $p^*$ : turning the surplus into a lower cap raises the price, so that in general equilibrium the equivalent rights are slightly greater than  $e_i^A - b_i / p^*$ . Therefore, the attainable emissions cut differs from  $\sum_i N_i b_i / p^*$ , which is why we solve for it numerically in Section 5.

Proposition 2 is a first-best statement. It abstracts from the second-best rationales for differentiation reviewed in Section 2. None of these requires the absence of transfers. What it questions is the argument that differentiation is needed because transfers would be infeasible: differentiation involves the same distributive outcome as well-chosen transfers; it only makes them implicit.

### 3.2 Dynamic setting

With several periods  $t$ , prices  $p_{it}$ , rights and emissions become trajectories, populations  $N_{it}$  change over time, and  $\beta_t$  denotes the discount factor. Under total utilitarianism, country  $i$ 's objective is  $\sum_t \beta_t N_{it} u(c_{it})$ . It is therefore convenient to state the dynamic results in totals rather than per capita:  $R_{it} \equiv N_{it} r_{it}$  and  $E_{it} \equiv N_{it} e_{it}$  are country  $i$ 's rights and emissions in period  $t$ ,  $E_{it}^A$  its emissions under the differentiated regime, and  $E_t = \sum_i E_{it}$  world emissions, so that the population weights are carried by the quantities themselves.

**Proposition 3** (Dynamic equivalence). *Suppose abatement costs are separable over time. To first order, country  $i$  is indifferent between the differentiated regime and the uniform price path  $(p_t^*)$  with rights  $(R_{it})$  if and only if*

$$\sum_t \beta_t p_t^* (R_{it} - E_{it}^A) = 0. \quad (7)$$

*If rights are a constant multiple  $\rho_i$  of an equal per capita share,  $r_{it} = \rho_i \bar{e}_t$ , the equivalent ratio is, to first order,*

$$\hat{\rho}_i = \frac{\sum_t \beta_t p_t^* N_{it} e_{it}^A}{\sum_t \beta_t p_t^* N_{it} \bar{e}_t} : \quad (8)$$

*the country's emissions under differentiation over its population share of world emissions, each period weighted by the discounted price.*

**Corollary 1** (Hotelling case). *If the discounted price  $\beta_t p_t^*$  is constant over time, the dynamic equivalence (7) only constrains cumulative rights  $R_i \equiv \sum_t R_{it}$ , which must equal the country's carbon budget under the differentiated regime, and which admits a first-order approximation in function of the price ratio  $\pi_i$  analogous to the static case if differentiated prices are homothetic,  $p_{it} = \pi_i p_t^*$ :*

$$R_i = E_i^A \approx E_i^* \left[ 1 + \varepsilon_i (1 - \pi_i) \right], \quad (9)$$

*where  $E_i^* \equiv \sum_t E_{it}^*$  (resp.  $E_i^A \equiv \sum_t E_{it}^A$ ) denotes  $i$ 's cumulative emissions under the uniform price (resp. autarky), and  $\varepsilon_i \equiv \sum_t \varepsilon_{it} E_{it}^* / E_i^*$  denotes the average over time, weighted by these emissions, of its price elasticities  $\varepsilon_{it}$  at  $p_t^*$ . Consequently,*

$$\hat{\rho}_i = \frac{E_i^A}{\sum_t \frac{N_{it}}{N_t} E_t^A}, \quad (10)$$

*that is the ratio of country  $i$ 's share of the cumulative emissions under differentiation to its equal per capita share of the world carbon budget.*

*Proof.* Apply the first-order argument of Proposition 1 period by period. In period  $t$  the consumption gain per capita is  $p_t^* (r_{it} - e_{it}^A)$  to first order; under total utilitarianism it enters the country's objective multiplied by its population, which turns per capita quantities into totals,  $N_{it}(r_{it} - e_{it}^A) = R_{it} - E_{it}^A$ , and it is discounted by  $\beta_t$ . Summing over  $t$  and setting the total to zero gives (7). Substituting  $R_{it} = \rho_i (N_{it}/N_t) E_t$  into (7) gives (8), and  $\beta_t p_t^*$  cancels from it when it is constant, leaving (10). Likewise, a constant  $\beta_t p_t^*$  factors out of (7), which then reads  $\sum_t (R_{it} - E_{it}^A) = 0$ . If  $p_{it} = \pi_i p_t^*$ , then (4) applied in each period gives  $E_{it}^A \approx E_{it}^* [1 + \varepsilon_{it} (1 - \pi_i)]$ , and summing over  $t$  gives (9).  $\square$

Equation (7) shows that timing matters. A tonne of rights is worth more in a period of high (discounted) prices, so the allocation depends on the whole price path, and a country’s cumulative carbon budget does not by itself determine its welfare. Only under a Hotelling price path, the natural benchmark with a fixed carbon budget and efficient capital markets, does the price weighting drop out, and (9) then states that a differentiated price is worth to a country the domestic carbon budget it induces.

## 4 Equivalent rights by country

### 4.1 Model

We use NICE2020 (Young-Brun et al. 2025), a recursive-dynamic integrated assessment model in the RICE tradition. It represents 179 countries, each disaggregated into income deciles, and is coupled to the FaIR climate model. Damages are country-specific: the damage function is applied to local temperatures. Abatement costs are not: all countries share the same cost function of the abatement rate and differ only through their baseline emissions. A given price therefore induces the same proportional abatement everywhere, so that the price elasticity  $\varepsilon_i$  of Proposition 1 is common to all countries at a given price, and cross-country differences in equivalent rights stem from differences in emissions and in prices. Emissions are measured on a consumption basis, so that a country’s allocation is a claim on its carbon footprint rather than on its territorial emissions. The distribution of both carbon costs and climate damages across deciles follows estimated income elasticities.

We measure a country’s welfare by the net present value over 2030–2100, discounted at 3% a year, of its population-weighted equally-distributed-equivalent (EDE) consumption (with inequality aversion  $\eta = 1.5$ ): the level of consumption which, equally distributed, would yield the same utilitarian welfare as the actual distribution of consumption across deciles. We also report the net present value of total consumption, which ignores within-country inequality. Domestic carbon revenues are recycled within each country in proportion to the inverse of marginal utility, that is using Negishi weights. This rule roughly preserves within-country inequality, so that our comparisons isolate between-country effects rather than the redistributive effect of recycling revenues through, say, a per capita dividend. International transfers enter national income and therefore affect investment and subsequent growth. Online Appendix A gives further details.

## 4.2 Design

The benchmark is an exponential uniform price path  $p_t^*$ , ramped up linearly from zero in 2025 and growing at a constant rate from 2031, which exhausts a carbon budget of 1000 GtCO<sub>2</sub> over 2025–2100. Among such paths, we take the one that maximises discounted world welfare, damages included, over 2030–2300.<sup>2</sup> The exponential path starts at \$127/t in 2030 and grows at 1.7% a year, reaching \$174/t in 2050 and \$339/t in 2090. World emissions fall from 38 GtCO<sub>2</sub> in 2025 to 24 in 2030 and 5 in 2090, and warming reaches 1.84°C in 2100. The price grows more slowly than the 3% discount rate, so that the discounted price falls over time: with damages in the objective, a tonne abated early is worth more than the carbon budget alone would imply. For a given country  $i$ , we compare two scenarios with the same global emission path.

- *Autarky*: country  $i$  applies  $\pi_i p_t^*$  and recycles its revenue domestically. The rest of the world applies a uniform price that we solve for year by year so that global emissions stay on the path of the uniform-price run. We allow country  $i$ 's abatement to exceed 100% of its emissions, which represents carbon dioxide removal, by extrapolating the cost curve beyond the backstop.
- *Uniform price*: all countries apply  $p_t^*$ . Country  $i$  receives rights per capita  $\rho_i$  times its population share of global emissions in the same year,  $r_{it} = \rho_i \bar{e}_t$ , and the remaining rights go to the rest of the world in proportion to each country's emissions.

A grid over  $(\pi_i, \rho_i)$  gives the net gain in welfare between the two scenarios, and hence country  $i$ 's indifference curve. We run the grid for eight diverse (sets of) countries.

## 4.3 Results

Figure 1 maps each country's welfare gain in the uniform-price regime relative to autarky over the  $(\pi_i, \rho_i)$  plane. Its indifference curve gives, for each autarky price factor, the smallest rights allocation at which the country prefers the uniform price.

**Current emissions poorly predict equivalent rights.** The intercept of each curve at  $\pi_i = 1$ , denoted  $\rho_1$ , is consistent with the theory: when the autarky price equals the uniform price, the equivalent rights match the country's cumulative

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<sup>2</sup>Welfare is discounted at a pure rate of time preference of 0.3%, which, with  $\eta = 1.5$  and the model's 1.8% annual growth of consumption per capita, corresponds to the 3% discount rate on consumption used throughout.

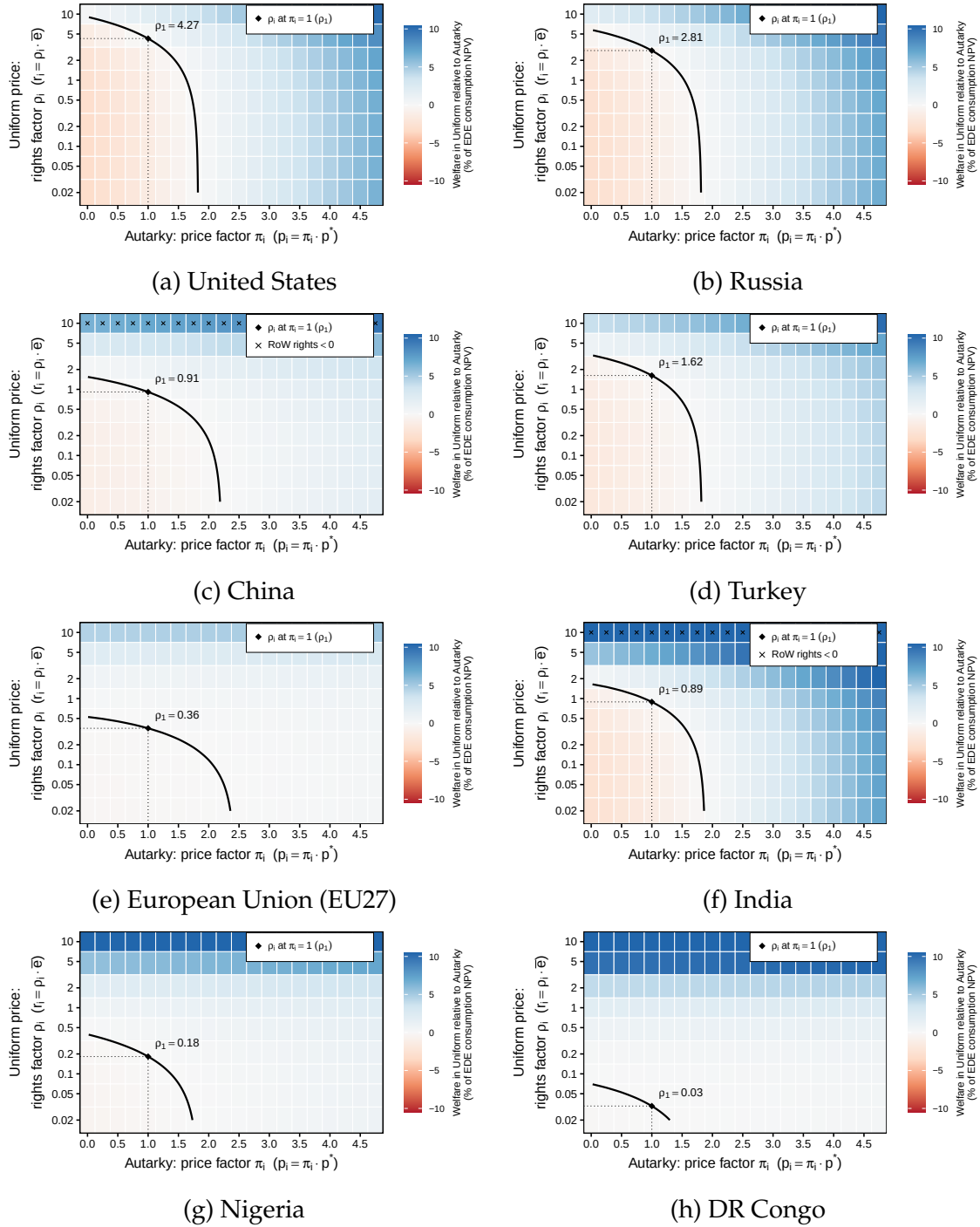


Figure 1: Welfare gain of the uniform-price regime over autarky

*Note:* The horizontal axis is country  $i$ 's autarky price as a multiple  $\pi_i$  of the benchmark price  $p_i^*$ , from 0 (no domestic carbon price) to 4.75, in steps of 0.25. The vertical axis (log scale) is its allocation  $\rho_i$  under the uniform price, as a multiple of world average emissions per capita. Both axes are the same in every panel. Colours give the difference in the NPV of EDE consumption over 2030–2100, in percent, on a common scale across countries clipped at the 95th percentile of the absolute difference ( $\pm 10.4$ ); blue cells favour the uniform price. The black line is the indifference curve and  $\rho_1$  its value at  $\pi_i = 1$ . Global emissions are the same in both regimes at every point of the grid, to within 0.01%. Crosses mark allocations that would leave the rest of the world with negative rights. The curve leaves the panel where the equivalent allocation turns negative, that is, where the country prefers the uniform price even while paying a net fee for its rights.

Table 1: Allocation at which each country is indifferent when the autarky price equals the benchmark ( $\pi_i = 1$ )

|               | Emissions p.c.<br>2030 (tCO <sub>2</sub> ) | Emissions p.c.<br>over world average | Predicted<br>$\hat{\rho}_1$ | Simulated<br>$\rho_1$ |
|---------------|--------------------------------------------|--------------------------------------|-----------------------------|-----------------------|
| United States | 10.87                                      | 3.83                                 | 4.27                        | 4.27                  |
| Russia        | 6.36                                       | 2.24                                 | 2.83                        | 2.81                  |
| China         | 4.88                                       | 1.72                                 | 0.93                        | 0.91                  |
| Turkey        | 3.21                                       | 1.13                                 | 1.64                        | 1.62                  |
| EU27          | 3.06                                       | 1.08                                 | 0.34                        | 0.36                  |
| India         | 1.32                                       | 0.47                                 | 0.94                        | 0.89                  |
| Nigeria       | 0.35                                       | 0.12                                 | 0.18                        | 0.18                  |
| DR Congo      | 0.03                                       | 0.01                                 | 0.03                        | 0.03                  |

*Note:*  $\hat{\rho}_1$  is equation (8) evaluated on the uniform-price run: the country's emissions over its population share of world emissions, each year weighted by the discounted benchmark price over 2030–2100.  $\rho_1$  is the intercept of the simulated indifference curve in Figure 1. The first two columns are 2030 snapshots, shown for reference.

emissions. It is worth noting that the equivalent rights account for future projected emissions while a 2025 snapshot does not. In 2030, China emits 1.7 times the world average per capita, yet its equivalent allocation is below an equal per capita share ( $\rho_1 = 0.91$ ), because its emissions fall steeply while the carbon price rises. Conversely, India emits less than half the world average in 2030 but is equivalent to 0.89 shares, because its emissions hold up. The EU27, at the world average in 2030, is equivalent to only 0.36 shares 1.

**Fair allocations cannot be replicated by differentiated prices.** While every price schedule has an equivalent allocation, not every allocation has an equivalent non-negative price. At a zero price, a country cannot emit more than its unabated emissions, so the most that differentiation can offer it is the allocation  $\rho_i(0)$  at which its curve meets the vertical axis. For low emitters, that bound lies far below an equal per capita share: 0.40 for Nigeria and 0.07 for the DRC. Burden-sharing principles that give such countries an equal per capita share or more — as historical responsibility or capacity to pay would — can only be achieved through a cap-and-trade or through monetary transfers.

## 5 Equivalent rights of prominent differentiation proposals

### 5.1 Proposals

We feed three proposals into the model as country-level price schedules.

[Wolfram et al. \(2025\)](#) imagine a coalition of 47 countries<sup>3</sup> including China, the EU, the UK, India and Brazil, which together emit 48% of world CO<sub>2</sub> in 2030. Members apply \$75, \$50 and \$25/t in high-, upper-middle- and lower-income countries, the tiers of the IMF’s international carbon price floor ([Parry et al. 2021](#)). The proposal targets four emissions-intensive industries, whereas NICE2020 has no sectoral detail. We apply the floors economy-wide, so our scenario is more ambitious than the one the authors model.

[Banerjee et al. \(2025\)](#) propose \$10, \$30 and \$50/t in low-, lower-middle- and upper-middle-income countries, in exchange for the compensation of climate damages, with revenues retained domestically. High-income countries fund the compensation and are expected to price carbon heavily, but the proposal gives no figure. Rather than leaving them unpriced, we price them at \$75/t, the top tier of [Wolfram et al. \(2025\)](#).

[Equal Right \(2023\)](#) proposes a system to phase out fossil fuel extraction that can be modeled as a schedule of differentiated prices. Each country sells its extraction licences at a price graduated by income group (\$240, \$120, \$60 and \$30/t) and discounted by up to 60% for climate vulnerability, which gives floors ranging from \$12 to \$240/t in 2025 (\$144/t on average), rising by about 16% a year to stay within 1.6°C. The revenues are pooled in a global fund that pays climate grants and a universal dividend. We use its schedule of price floors, which is far more dispersed than the other two, together with its global fund: the revenue of every country is pooled and returned as an equal per capita dividend to the world population. Equal Right is therefore the one schedule of the three whose explicit transfers we model, and the only one whose revenue leaves the country that raises it. The report sets a charge for 173 of the 179 economies of the model; we price the remaining six at the median charge of their World Bank income group, so that the schedule covers the world and its club is every country.

The first two proposals do not specify how prices evolve over time. [Parry et al. \(2021\)](#) set their floors for 2030, [Wolfram et al. \(2025\)](#) simulate a single illustrative year and suggest “sunset provisions that gradually raise lower-tier prices so countries ‘graduate’ into a single uniform price”, and [Banerjee et al. \(2025\)](#) raise a country’s price only when it moves to a higher income group. For each

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<sup>3</sup>The members are Algeria, Australia, Brazil, Cameroon, Canada, China, Egypt, the European Union, Ghana, Iceland, India, Indonesia, Kenya, Liechtenstein, Mozambique, Norway, Switzerland, Thailand, Togo, the United Kingdom, Uganda and Zambia.

proposal, we hold the announced prices from 2025 through 2030 and raise them by 5% a year thereafter, capped at the backstop price. While [Wolfram et al. \(2025\)](#) and [Banerjee et al. \(2025\)](#) include unquantified provisions to make prices converge, we do not model them to properly compare differentiated prices with a cap-and-trade. In the Online Appendix, we also run Equal Right’s original escalation path, which reaches the backstop in most countries by 2050. The proposals yield a 2100 warming of 2.16°C ([Wolfram et al. 2025](#)), 1.80°C ([Banerjee et al. 2025](#)) and 1.66°C (Equal Right).

## 5.2 Design

We compare each differentiation proposal with a cap-and-trade. Its members are countries the proposal prices, and they face a uniform price  $p_t^*$ , calibrated so that their joint emissions equal those under differentiation. This price is \$49.2/t ([Wolfram et al. 2025](#)), \$55.8/t ([Banerjee et al. 2025](#)) and \$138.5/t (Equal Right) in 2030, and then grows by 4–5% a year for the first two. Non-members are unpriced in both regimes. Member  $i$  receives rights per capita  $r_{it} = \rho_i \bar{e}_t$ , where  $\bar{e}_t$  is the club’s emissions per capita under the proposal, so that  $\rho_i = 1$  is an equal per capita allocation.

We solve for the equivalent ratios  $\rho_i^*$  on two criteria: welfare (the NPV of population-weighted EDE consumption) or (the NPV of) total consumption. Proposition 2 implies that the club regime leaves a surplus, which we use in either of two ways:

- *Reduced emissions:* rights are cut (hence the price and abatement costs increased) until every member is as well off as under the proposal. As in Propositions 1 and 2, members are compared at the same damages: while solving for the allocation, we hold each country’s damages at the level implied by the proposal’s own temperature path. This is an accounting device used to isolate the efficiency gain of uniform pricing. In case the price is below the social cost of carbon, cutting rights actually improves welfare, so accounting for avoided damages would tighten the cap until the price reaches a point above the social cost of carbon. Reduced emissions would then measure the ambition of a proposal rather than the instrument used to deliver it (Online Appendix B). Once rights cuts are computed, we reintegrate avoided damages into the welfare and consumption gains we report.
- *Increased consumption:* joint emissions are kept at their level under the proposal in every year, and the surplus rights are shared so as to maximise the smallest relative gain of any member, which equalises the gains of all members.

We solve each problem jointly for all members. At each iteration, the allocation sets the cap, the club price is recalibrated to deliver it, and each member's ratio is updated until convergence. In the Online Appendix, we also report an *isolated* solve, which finds each country's ratio in turn while every other member is granted its own emissions under the proposal. It is the partial-equilibrium analogue of Proposition 1, but it credits each country with the whole efficiency gain of joining the market, a gain that cannot be paid to all members at once. With the isolated solve, the surplus is taken either as reduced emissions, by applying the isolated ratios together, or as increased consumption, by scaling them up uniformly to the proposal's emissions. The Online Appendix also reports two alternative ways of sharing the surplus rights (using either the *joint* or *isolated* solve): in proportion to population times marginal utility, and by uniform scaling.

### 5.3 Results

Table 2 reports, for ten major economies, the allocation equivalent to each proposal, solved on welfare and on consumption, together with the outcome of the two ways of using the surplus.

Table 2: Equivalent rights of three differentiated-price schedules, and the dividend of a uniform price

| Country                                                                  | Wolfram et al. |                      |                      | Banerjee, Duflo & Greenstone |              |                      |                      | Equal Right, 5%/yr |                      |                      |
|--------------------------------------------------------------------------|----------------|----------------------|----------------------|------------------------------|--------------|----------------------|----------------------|--------------------|----------------------|----------------------|
|                                                                          | $p$            | $\rho^{\text{welf}}$ | $\rho^{\text{cons}}$ | $p$                          | $\hat{\rho}$ | $\rho^{\text{welf}}$ | $\rho^{\text{cons}}$ | $p$                | $\rho^{\text{welf}}$ | $\rho^{\text{cons}}$ |
| United States                                                            | 0              | –                    | –                    | 75                           | 3.0          | 2.46                 | 2.92                 | 240                | –                    | –                    |
| Russia                                                                   | 0              | –                    | –                    | 75                           | 1.98         | 1.61                 | 1.88                 | 240                | –                    | –                    |
| European Union                                                           | 75             | 0.31                 | 0.33                 | 75                           | 0.26         | 0.28                 | 0.29                 | 232                | –                    | –                    |
| Turkey                                                                   | 0              | –                    | –                    | 50                           | 1.57         | 1.61                 | 1.55                 | 108                | –                    | –                    |
| Indonesia                                                                | 50             | 0.96                 | 1.01                 | 50                           | 1.12         | 1.13                 | 1.1                  | 96                 | –                    | –                    |
| China                                                                    | 50             | 0.97                 | 0.98                 | 50                           | 0.9          | 0.94                 | 0.92                 | 108                | –                    | –                    |
| Brazil                                                                   | 50             | 0.45                 | 0.47                 | 50                           | 0.49         | 0.53                 | 0.49                 | 108                | –                    | –                    |
| India                                                                    | 25             | 1.25                 | 1.21                 | 30                           | 1.28         | 1.18                 | 1.13                 | 42                 | –                    | –                    |
| Nigeria                                                                  | 0              | –                    | –                    | 30                           | 0.26         | 0.25                 | 0.23                 | 42                 | –                    | –                    |
| DR Congo                                                                 | 0              | –                    | –                    | 10                           | 0.1          | 0.07                 | 0.06                 | 15                 | –                    | –                    |
| <i>Reduced emissions: every member as well off as under the proposal</i> |                |                      |                      |                              |              |                      |                      |                    |                      |                      |
| World temp. 2100, change (°C)                                            |                | –0.010               | –0.010               |                              |              | –0.017               | –0.019               |                    | –                    | –                    |
| Emissions change in the coalition (%)                                    |                | –4.9%                | –5.1%                |                              |              | –4.3%                | –4.7%                |                    | –                    | –                    |
| World welfare gain (%)                                                   |                | 0.041%               | 0.034%               |                              |              | 0.112%               | 0.074%               |                    | –                    | –                    |
| World consumption gain (%)                                               |                | 0.028%               | 0.027%               |                              |              | 0.044%               | 0.040%               |                    | –                    | –                    |
| Members losing on welfare                                                |                | 0                    | 1                    |                              |              | 1                    | 36                   |                    | –                    | –                    |
| Members losing on consumption                                            |                | 12                   | 0                    |                              |              | 34                   | 1                    |                    | –                    | –                    |
| <i>Increased consumption: coalition emissions as under the proposal</i>  |                |                      |                      |                              |              |                      |                      |                    |                      |                      |
| World welfare gain (%)                                                   |                | 0.015%               | 0.006%               |                              |              | 0.104%               | 0.054%               |                    | –                    | –                    |
| World consumption gain (%)                                               |                | 0.020%               | 0.019%               |                              |              | 0.057%               | 0.053%               |                    | –                    | –                    |
| Smallest member gain, welfare                                            |                | 0.046%               | –0.022%              |                              |              | 0.056%               | –0.174%              |                    | –                    | –                    |
| Smallest member gain, consumption                                        |                | –0.099%              | 0.038%               |                              |              | –0.381%              | 0.052%               |                    | –                    | –                    |
| Members losing on welfare                                                |                | 0                    | 2                    |                              |              | 0                    | 36                   |                    | –                    | –                    |
| Members losing on consumption                                            |                | 6                    | 0                    |                              |              | 23                   | 0                    |                    | –                    | –                    |

*Note:* Joint solve.  $p_{2030}$  is the price the schedule asks of the country (\$/t); the comparison regime prices every member of the schedule’s own club at one common price instead (\$49.2/t, \$55.8/t and \$138.5/t respectively in 2030, growing by 4–5% a year), and leaves non-members unpriced, as the schedule does; of the three, only the [Wolfram et al. \(2025\)](#) schedule has non-members, the other two pricing every country.  $\rho$  is the country’s allocation as a multiple of the club’s emissions per capita under the schedule, so that  $\rho = 1$  is an equal per capita allocation;  $\rho^{\text{welf}}$  and  $\rho^{\text{cons}}$  hold the country at, respectively, its welfare (NPV of equally-distributed-equivalent consumption) and its total consumption under the schedule, both population-weighted.  $\hat{\rho}$  is the first-order prediction of equation (8): the country’s emissions under the schedule over its population share of the club’s, each year weighted by the discounted club price. A country that its schedule does not price is outside that club in both regimes ( $p = 0$ , no allocation). Under the two schedules of [Wolfram et al. \(2025\)](#) and [Banerjee et al. \(2025\)](#) each country keeps its own revenue; under Equal Right it is pooled worldwide and returned as an equal per capita dividend, as that proposal specifies, which is why its equivalent allocations are so much lower for large per capita emitters and can even be negative. The schedules themselves reach 2.16°C, 1.80°C and 1.66°C in 2100. Under *reduced emissions*, rights are cut until every member is as well off as under the schedule, holding each country’s damages at the level implied by the schedule’s temperature path, so that the cut measures the efficiency gain of uniform pricing and not the avoided damages it brings; the temperature and world rows are nonetheless computed with damages endogenous, and so include those avoided damages. Under *increased consumption*, the club’s emissions are held at their level under the schedule and the surplus is shared so as to equalise members’ relative gains. A member counts as losing when it falls short of its level under the schedule by more than 0.005%, four times the tolerance of the solve.

**The surplus as reduced emissions.** The uniform-price equivalent allocation of the Banerjee et al. differentiation reduces global emissions by 4.7% and its 2100 warming by 0.02°C, while preserving pre-damage consumption in every country. Reduced warming makes every country better off than under differentiation except for Mongolia, which would actually benefit from extra warming, according to our model (Online Appendix B). The corresponding emission cuts are 5.1% for Wolfram et al. (2025) and 15.2% for the Equal Right schedule.

**The surplus as increased consumption.** The alternative is to keep total emissions at their level under differentiation and to share the surplus rights so as to raise the smallest gain, which equalises the gains of all members. Every country then gains 0.052% of the net present value of its consumption vis-à-vis Banerjee et al.’s proposal.

**The equivalent rights.** Under the schedule of Banerjee et al. (2025), the equivalent rights range from 2.92 equal per capita shares for the United States to 0.06 for the DRC.

The first-order prediction  $\hat{\rho}_i$  of equation (8), shown for the Banerjee et al. schedule, reproduces the solved allocations closely. As Proposition 1 implies, it errs on the generous side for most countries, and by most for those priced furthest below the uniform price, whose second-order term is largest: 1.28 against 1.13 for India, 0.10 against 0.06 for the DRC.

The equivalent rights track the two determinants of Proposition 1: a country’s emissions, and the gap between the price it is asked and the club price of \$56/t. The United States and Russia, priced at \$75/t and emitting well above the world average, need 2.92 and 1.88 shares; China, priced at \$50/t, needs 0.92; India, at \$30/t, 1.13; Nigeria, at the same price but emitting a tenth as much per capita, 0.23. The schedule of Wolfram et al. (2025) gives a similar picture among its 47 members. The Equal Right schedule, whose revenue does not stay at home, is a different matter.

**A schedule with a global dividend.** Under Equal Right the revenue of every country is pooled and paid back as an equal per capita dividend, so the equivalent allocation has to replace what the schedule charges a country *and* what the fund pays it. To first order, a member is indifferent when  $p_t^* R_{it} = p_t^* E_{it}^A + D_t N_{it}/N_t - p_{it} E_{it}^A$ , with  $D_t = \sum_j p_{jt} E_{jt}^A$  the world revenue and  $p_{it} E_{it}^A$  foregone revenue from domestic pricing. Denoting  $\bar{p}_t \equiv D_t/E_t$  is the average price paid under differentiation, the dividend is therefore worth not one equal per capita share but  $\bar{p}/p^*$  of one, about 0.72 under this schedule. Indeed, the members charged \$240/t abate almost everything, so the tonnes that survive sit in the \$15–42/t tiers, and the emission-weighted mean price falls far below the

uniform price that delivers the same abatement. A member that emits almost nothing therefore ends near  $\bar{p}/p^*$  rather than at one: this explains the equivalent right ratio below unity for a country such as the DRC. For ten countries, the equivalent ratio is negative:  $-0.35$  for the United States and  $-0.04$  for Russia, meaning that these countries would have a net duty to remove carbon rather than a net right to emit, in line with the fairness principle accounting for historical responsibility (Fanning & Hickel 2023).

## 6 Conclusion

Differentiating carbon prices by income is increasingly proposed as a way to reconcile carbon pricing with equity. Yet a differentiated schedule is equivalent, to first order, to a uniform price with tradable rights that grandfather each country on the emissions the schedule itself induces. These are implicit transfers of the same first-order size, only paid in (excess or forgone) abatement rather than in money. Moreover, since abatement costs are convex, the explicit version leaves every country better off.

Replacing the differentiated prices of Wolfram et al. (2025) or Banerjee et al. (2025) with a uniform price at equivalent rights would cut the coalition's emissions by about 5% while leaving every member as well off as under differentiation, and by 15% under the more dispersed schedule of Equal Right (2023). Alternatively, at unchanged emissions, it would raise every member's consumption.

The equivalence runs only one way. Every schedule of prices has an equivalent allocation of rights, but no schedule of non-negative prices can deliver to a low emitter what a generous allocation would: even a zero price leaves Nigeria below two fifths of an equal per capita share, and the DRC below a tenth. Allocations based on equal per capita rights, historical responsibility or capacity to pay are therefore out of reach of price differentiation.

Two implications follow for negotiations. First, the equivalence formula provides a linear correspondence between a price and emission rights, allowing parties to shift the discussion from one notion to the other as needed. Second, the value of national sovereignty invoked to dismiss an international cap-and-trade should be balanced with the efficiency gains that an international commitment would enable. Furthermore, survey evidence shows that 85% of worldwide respondents state that climate policy at the *global* level is needed, at odd with the supposed taste for national sovereignty (Fabre et al. 2025). Finally, robust majority acceptance of an international cap-and-trade funding equal per capita cash transfers indicates that most people are ready to accept large North–South transfers that differentiated autarky prices cannot deliver.

The analysis has limitations. NICE2020 has capital market, so it cannot cap-

ture the diverging capital return motive for differentiation. It has no sectoral detail, which forced us to apply sectoral proposals economy-wide and ignores the country-specific costs of sectoral reallocation. Extending the equivalence to settings with trade and different sectors, where a uniform price must coexist with border adjustments and territorial emissions can be distinguished from carbon footprints, is a natural next step to better inform climate negotiators.

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## Declaration of competing interest

The authors declare no competing interests.

## Data availability

The code (Julia, built on the Mimi implementation of NICE2020) and all inputs needed to reproduce the results are publicly available on [github.com/bixiou/NICE2020](https://github.com/bixiou/NICE2020).

## Declaration of generative AI use

During the preparation of this work the authors used Claude Code (Opus 5) to assist with code and editing. The authors reviewed and edited the content and take full responsibility for it.

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# Online Appendix

## A Model details

**Configuration.** All runs use NICE2020 with revenue recycling within each country in proportion to  $c^{\eta}$ , computed once from a uniform-price reference run and held fixed across runs (under the Equal Right schedule the revenue is first pooled worldwide and paid out as an equal per capita dividend, as the proposal specifies, and the  $c^{\eta}$  rule then shares each country's dividend across its deciles), the default income elasticity of emissions across deciles, emissions measured on a consumption basis (carbon footprints), and international transfers entering national income, hence investment. The same configuration is used in the differentiated and in the uniform-price regime, so that the two regimes differ only in prices and rights.

**Rest-of-world price in the autarky scenario.** When country  $i$  applies  $\pi_i p_t^*$ , the rest of the world applies a price  $p_{-i,t}$  such that global emissions follow the path of the uniform-price run. Since every country abates at the rate  $\mu = (p/p_t^{\text{back}})^{1/(\theta_2-1)}$ , with  $\theta_2 = 2.6$ , the rest of the world's emissions are  $B_{-i,t}(1 - \mu_{-i,t})$ , where  $B_{-i,t}$  is its baseline. We therefore update  $p_{-i,t}$  by Newton steps in the abatement rate,  $\mu_{-i,t} \leftarrow 1 - (E_t - E_{it})/B_{-i,t}$ , re-evaluating  $B_{-i,t}$  in each run, until world emissions are within 0.01% of the target in every year.

**Rights of the rest of the world in the uniform-price scenario.** Country  $i$  receives  $\rho_i$  times its population share of global emissions in each year. The remainder is distributed to other countries in proportion to their emissions, which keeps total rights equal to the cap for any  $\rho_i$ . When country  $i$ 's rights exceed the cap, the remainder is negative and the rest of the world pays for its emissions.

**Abatement beyond 100%.** Standard implementations constrain the abatement rate to  $\mu \in [0, 1]$ . Under high autarky prices, country  $i$ 's price exceeds the backstop price and this constraint binds. In the autarky scenario we extrapolate the cost function beyond  $\mu = 1$ , which represents carbon dioxide removal at rising marginal cost. The proposals of Section 5 are instead capped at the backstop price.

**Viability flags.** We flag grid points at which (i) the rest-of-world price has to be negative in some year to keep the global path, and (ii) country  $i$ 's rights exceed global emissions in some year, leaving the rest of the world with negative rights. Only the second occurs, for China and India at allocations of six equal per

capita shares or more (the crosses in Figure 1). The first never does: the lowest rest-of-world price over the whole grid (after 2025, when the benchmark itself is still zero) is \$3.4/t, for China at  $\pi_i = 4.75$  in 2026. Even when a large emitter prices carbon at nearly five times the benchmark, holding world emissions on the benchmark path only requires the rest of the world to price carbon less, never to pay for emissions. The figures this replaces flagged such points for some countries, but on a different quantity: the condition was evaluated on the linear first guess for the rest-of-world price, whose emission shares are divided by a world total that approaches zero late in the century.

Table A1: Autarky scenario for the United States ( $\pi_i = 2$ )

|                                  | 2030  | 2040  | 2050  | 2060  | 2070  | 2080  | 2090  | 2100  |
|----------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| $p^*$ (USD/tCO <sub>2</sub> )    | 127.0 | 147.6 | 174.3 | 205.9 | 243.2 | 287.3 | 339.4 | 401.0 |
| $p_i$ (USD/tCO <sub>2</sub> )    | 254.0 | 295.1 | 348.6 | 411.8 | 486.5 | 574.7 | 678.9 | 801.9 |
| $p_{-i}$ (USD/tCO <sub>2</sub> ) | 106.9 | 121.2 | 140.5 | 167.9 | 202.7 | 245.7 | 297.0 | 358.1 |
| $E_i$ (GtCO <sub>2</sub> )       | 2.55  | 1.94  | 1.14  | 0.55  | −0.00 | −0.49 | −0.86 | −1.11 |

Table A2: Uniform-price scenario for the Democratic Republic of Congo ( $\rho_i = 0.5$ )

|                               | 2030  | 2040  | 2050  | 2060  | 2070  | 2080  | 2090  | 2100  |
|-------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| $p^*$ (USD/tCO <sub>2</sub> ) | 127.0 | 147.6 | 174.3 | 205.9 | 243.2 | 287.3 | 339.4 | 401.0 |
| $R_i$ (GtCO <sub>2</sub> )    | 0.170 | 0.165 | 0.149 | 0.145 | 0.134 | 0.114 | 0.082 | 0.041 |
| $E_i$ (GtCO <sub>2</sub> )    | 0.004 | 0.006 | 0.009 | 0.012 | 0.014 | 0.013 | 0.010 | 0.005 |
| Transfer (bn USD)             | 21.1  | 23.5  | 24.4  | 27.3  | 29.3  | 29.1  | 24.6  | 14.6  |

Note:  $R_i$  and  $E_i$  are the country's rights and emissions (GtCO<sub>2</sub>), and the transfer is the value of its net sales of rights,  $(R_i - E_i) p^*$ .

## B Solves, sharing rules and numerical accuracy

**The two solves.** The *joint* solve of the main text finds all  $\rho_i$  at once. At each iteration the allocation sets the cap, the club price is recalibrated to deliver it, and each member's ratio is moved by a secant step on its gap to the target. For the reduced-emissions variant, an outer loop then moves the level of the allocation until the members' common gain is zero, so that every member ends exactly at its level under the schedule. The *isolated* solve instead finds each country's ratio in turn, with every other member granted its own emissions under the schedule and the club price held at the value that reproduces the schedule's emissions. It is the

partial-equilibrium counterpart of Proposition 1: it ignores that the other members also change their allocation, and that the club price adjusts to the resulting cap.

The two solves agree on the aggregate. Taking the equivalent allocation at face value cuts the Wolfram et al. club's emissions by 6.9% (isolated) against 6.9% (joint), and the Banerjee et al. club's by 6.7% against 6.0%. They differ on the distribution, and the difference is systematic: the isolated solve credits each country in turn with the whole efficiency gain of joining the market, a gain that cannot be paid to every member at once. Applied together, the isolated allocations leave 25 of 47 Wolfram et al. members and 31 of 179 Banerjee et al. members below their level under the schedule, while the joint allocation leaves none. Table A3 reports the isolated allocations in the format of Table 2.

**Holding damages fixed.** In the reduced-emissions variant, the cap is tighter than the schedule's, so warming is lower and every country's damages fall. Propositions 1 and 2 compare the two regimes at the same damages, and so does our solve: when searching for the allocation that leaves each member as well off as under the schedule, we hold each country's damages at the level implied by the schedule's own temperature path. Emissions still fall, and the avoided damages are real; they are simply not allowed to pay for the cut that the solve is measuring. The device is the numerical counterpart of the statement, in Proposition 2, that the cap can be tightened by  $\sum_i N_i b_i / p^*$  tonnes with no country worse off: that statement too changes emissions while treating damages as given, and is to be read as a conversion of the surplus into tonnes rather than as a description of a world.

Letting damages respond instead answers a different question. The solve then keeps tightening the cap for as long as the club's avoided damages exceed its marginal abatement cost, that is, until the club price reaches the club's own social cost of carbon. Its answer is the club's optimal cap, which depends on the ambition of climate policy rather than on the instrument that delivers it; in a preliminary solve for Banerjee et al. (2025), whose club is the whole world, the cap had fallen by 21% and every member was still better off when the solve was stopped. We therefore do not report it. The avoided damages that the tighter cap does deliver are reported separately, as the fall in 2100 warming and as the world welfare gain of the variant, both computed with damages endogenous.

**Alternative sharing rules.** The increased-consumption variant of the main text shares the surplus so as to maximise the smallest relative gain, which equalises the gains of all members. Two other rules are reported in Table A4. Sharing the surplus in proportion to population times marginal utility directs it to the poorest members: it yields by far the largest inequality-averse gain (a world welfare gain

of 0.231% for Banerjee et al., against 0.054% under the maximin rule) but, unlike the maximin rule, it leaves 11 of 179 members below their consumption under the schedule. Uniform scaling of the whole allocation is dominated: the extra rights go to the largest holders, and the looser cap lowers the club price, which devalues the rights of small, generously endowed net sellers. It leaves no member below on consumption, but its inequality-averse gain is the smallest of the three rules (0.039% for Banerjee et al.), and 9 of 47 Wolfram et al. members and 64 of 179 Banerjee et al. members end below their welfare under the schedule.

**Who loses from the avoided damages.** The equivalence is solved with damages held at the schedule’s temperature path, and every member is then within 0.001% of indifference. The gains reported in Table 2 are computed with damages endogenous, so they also carry the avoided damages of the tighter cap, which are shared in proportion to each country’s marginal damage  $\beta_1 + 2\beta_2 T$ . In the Kalkuhl–Wenz specification of NICE2020, 13 countries have  $\beta_1 < 0$  and a local anomaly below  $-\beta_1/2\beta_2$ , so their marginal damage is negative: they gain from warming, and less of it makes them worse off. They are the only members that end below their level under the schedule — Mongolia under Banerjee et al. (2025), and Finland, Iceland, Mongolia and Russia under Equal Right at 5%/yr, ordered by  $|\beta_1|$  as the mechanism implies, and none under Wolfram et al. (2025), whose cap moves warming least. The loss is a mirror image of the gain: Nigeria and India, whose marginal damage is largest, gain 0.23% and 0.18% of their consumption from the same avoided warming, against Mongolia’s  $-0.05\%$ . This is a property of the damage function, not of the equivalence: at the damages of the schedule, on which the equivalence is defined, no member loses.

**Numerical accuracy.** The club price is calibrated so that members’ emissions match the cap within 0.01% of the club’s peak emissions in every year. The joint solves stop when no member’s gap to its target exceeds 0.002% of its level under the schedule, and the reduced-emissions outer loop when the common gain is below 0.001%. A member is counted as losing when it falls short by more than 0.005%, four times the tolerance. Prices apply from 2025, while welfare is the net present value over 2030–2100, so what happens in the five first years enters the objective only through its effect on capital accumulation. Measured on the Wolfram et al. allocation, this does not flatter the dividend: read over 2025–2100 instead, members are on average 0.029% *above* their level under the schedule, so the reported cut is, in the aggregate, conservative. It does affect the distribution across members, who range from  $-0.16\%$  to  $+0.14\%$  on that longer window against  $-0.0003\%$  to  $+0.0008\%$  on the window solved. An equivalence solved on 2025–2100 would therefore give the same aggregate dividend and a somewhat different allocation.

Table A3: Allocations equivalent to the three schedules — isolated solve

| Country                                                                  | Wolfram et al. |                      |                      | Banerjee, Duflo & Greenstone |           |                      |                      | Equal Right, 5%/yr |                      |                      |
|--------------------------------------------------------------------------|----------------|----------------------|----------------------|------------------------------|-----------|----------------------|----------------------|--------------------|----------------------|----------------------|
|                                                                          | $p$            | $\rho^{\text{welf}}$ | $\rho^{\text{cons}}$ | $p$                          | $\hat{p}$ | $\rho^{\text{welf}}$ | $\rho^{\text{cons}}$ | $p$                | $\rho^{\text{welf}}$ | $\rho^{\text{cons}}$ |
| United States                                                            | 0              | —                    | —                    | 75                           | 3.0       | 2.13                 | 2.78                 | 240                | −1.94                | −1.13                |
| Russia                                                                   | 0              | —                    | —                    | 75                           | 1.98      | 1.39                 | 1.78                 | 240                | −0.99                | −0.52                |
| European Union                                                           | 75             | 0.27                 | 0.29                 | 75                           | 0.26      | 0.27                 | 0.29                 | 232                | 0.46                 | 0.5                  |
| Turkey                                                                   | 0              | —                    | —                    | 50                           | 1.57      | 1.56                 | 1.55                 | 108                | 0.95                 | 0.9                  |
| Indonesia                                                                | 50             | 0.89                 | 1.0                  | 50                           | 1.12      | 1.1                  | 1.1                  | 96                 | 0.96                 | 0.92                 |
| China                                                                    | 50             | 0.88                 | 0.93                 | 50                           | 0.9       | 0.93                 | 0.92                 | 108                | 1.03                 | 0.98                 |
| Brazil                                                                   | 50             | 0.38                 | 0.46                 | 50                           | 0.49      | 0.5                  | 0.49                 | 108                | 0.85                 | 0.81                 |
| India                                                                    | 25             | 1.31                 | 1.28                 | 30                           | 1.28      | 1.19                 | 1.16                 | 42                 | 1.46                 | 1.36                 |
| Nigeria                                                                  | 0              | —                    | —                    | 30                           | 0.26      | 0.26                 | 0.23                 | 42                 | 0.88                 | 0.86                 |
| DR Congo                                                                 | 0              | —                    | —                    | 10                           | 0.1       | 0.08                 | 0.06                 | 15                 | 0.78                 | 0.78                 |
| <i>Reduced emissions: every member as well off as under the proposal</i> |                |                      |                      |                              |           |                      |                      |                    |                      |                      |
| World temp. 2100, change (°C)                                            |                | −0.016               | −0.010               |                              |           | −0.029               | −0.020               |                    | −0.046               | −0.044               |
| Emissions change in the coalition (%)                                    |                | −8.3%                | −5.3%                |                              |           | −7.3%                | −4.9%                |                    | −18.5%               | −17.7%               |
| World welfare gain (%)                                                   |                | 0.082%               | 0.046%               |                              |           | 0.182%               | 0.089%               |                    | 0.237%               | 0.143%               |
| World consumption gain (%)                                               |                | 0.036%               | 0.029%               |                              |           | 0.052%               | 0.042%               |                    | 0.107%               | 0.105%               |
| Members losing on welfare                                                |                | 30                   | 10                   |                              |           | 38                   | 41                   |                    | 81                   | 74                   |
| Members losing on consumption                                            |                | 36                   | 24                   |                              |           | 42                   | 33                   |                    | 82                   | 80                   |
| <i>Increased consumption: coalition emissions as under the proposal</i>  |                |                      |                      |                              |           |                      |                      |                    |                      |                      |
| World welfare gain (%)                                                   |                | 0.021%               | 0.010%               |                              |           | 0.100%               | 0.051%               |                    | 0.105%               | 0.018%               |
| World consumption gain (%)                                               |                | 0.021%               | 0.020%               |                              |           | 0.057%               | 0.053%               |                    | 0.182%               | 0.177%               |
| Smallest member gain, welfare                                            |                | −0.089%              | −0.107%              |                              |           | −0.188%              | −0.206%              |                    | −5.006%              | −5.019%              |
| Smallest member gain, consumption                                        |                | −0.124%              | −0.102%              |                              |           | −0.408%              | −0.163%              |                    | −7.390%              | −7.409%              |
| Members losing on welfare                                                |                | 4                    | 7                    |                              |           | 34                   | 61                   |                    | 65                   | 46                   |
| Members losing on consumption                                            |                | 18                   | 4                    |                              |           | 37                   | 28                   |                    | 68                   | 64                   |

Note: As Table 2, with each country's allocation solved on its own while every other member holds its emissions under the schedule and the club price is held at the value that reproduces the schedule's emissions.

Table A4: Club-price rights ratios  $\rho_i$  equivalent to the Wolfram et al. and Banerjee, Duflo & Greenstone proposals

| Country                                               | Wolfram et al. |                          |                       | Banerjee, Duflo & Greenstone |                          |                       |
|-------------------------------------------------------|----------------|--------------------------|-----------------------|------------------------------|--------------------------|-----------------------|
|                                                       | $p_i$          | $\rho^{\text{isolated}}$ | $\rho^{\text{joint}}$ | $p_i$                        | $\rho^{\text{isolated}}$ | $\rho^{\text{joint}}$ |
| United States                                         | 0              | –                        | –                     | 75                           | 2.78                     | 2.92                  |
| Russia                                                | 0              | –                        | –                     | 75                           | 1.78                     | 1.88                  |
| European Union                                        | 75             | 0.29                     | 0.33                  | 75                           | 0.29                     | 0.29                  |
| Turkey                                                | 0              | –                        | –                     | 50                           | 1.55                     | 1.55                  |
| Indonesia                                             | 50             | 1.0                      | 1.01                  | 50                           | 1.1                      | 1.1                   |
| China                                                 | 50             | 0.93                     | 0.98                  | 50                           | 0.92                     | 0.92                  |
| Brazil                                                | 50             | 0.46                     | 0.47                  | 50                           | 0.49                     | 0.49                  |
| India                                                 | 25             | 1.28                     | 1.21                  | 30                           | 1.16                     | 1.13                  |
| Nigeria                                               | 0              | –                        | –                     | 30                           | 0.23                     | 0.23                  |
| DR Congo                                              | 0              | –                        | –                     | 10                           | 0.06                     | 0.06                  |
| <i>Reduced emissions (variant 1)</i>                  |                |                          |                       |                              |                          |                       |
| Emissions change in the coalition (%)                 |                | –5.3%                    | –5.1%                 |                              | –4.9%                    | –4.7%                 |
| World temp. 2100, change (°C)                         |                | –0.010                   | –0.010                |                              | –0.020                   | –0.019                |
| World welfare gain (%)                                |                | 0.046%                   | 0.034%                |                              | 0.089%                   | 0.074%                |
| World consumption gain (%)                            |                | 0.029%                   | 0.027%                |                              | 0.042%                   | 0.040%                |
| <i>Increased consumption (variant 2)</i>              |                |                          |                       |                              |                          |                       |
| World welfare gain (%)                                |                | 0.010%                   | 0.006%                |                              | 0.051%                   | 0.054%                |
| World consumption gain (%)                            |                | 0.020%                   | 0.019%                |                              | 0.053%                   | 0.053%                |
| <i>Surplus shared by marginal utility (variant 3)</i> |                |                          |                       |                              |                          |                       |
| World welfare gain (%)                                |                | 0.043%                   | 0.033%                |                              | 0.246%                   | 0.231%                |
| World consumption gain (%)                            |                | 0.022%                   | 0.021%                |                              | 0.058%                   | 0.057%                |
| <i>Surplus shared by uniform scaling (variant 4)</i>  |                |                          |                       |                              |                          |                       |
| World welfare gain (%)                                |                | =(2)                     | 0.001%                |                              | =(2)                     | 0.039%                |
| World consumption gain (%)                            |                | =(2)                     | 0.018%                |                              | =(2)                     | 0.052%                |

Note:  $p_i$  is the carbon price the proposal asks of country  $i$  in 2030; the comparison regime instead prices every member of the proposal's own club at one common price (\$49.2/t for Wolfram et al. and \$55.8/t for Banerjee, Duflo & Greenstone in 2030) and leaves non-members unpriced, as the proposal does.  $\rho_i$  is the country's allocation as a multiple of an equal-per-capita share of the club's emissions under the proposal, taken at face value in variant 1;  $\rho^{\text{isolated}}$  solves each country on its own,  $\rho^{\text{joint}}$  all of them jointly. The European Union figure aggregates its members' rights, and a country the proposal does not price is outside the club in both regimes, shown at  $p_i = 0$  with no equivalent allocation ('–'). Variant 1 (reduced emissions) lets the equivalent allocation set the cap: in the joint solve every member is held exactly at its level under the proposal; in the isolated solve each country's ratio is solved with all others grandfathered, and the ratios are then applied together. Variants 2 to 4 (increased consumption) keep club emissions at the proposal's and share the surplus: equalising the relative gains of all members (joint variant 2), by scaling the isolated allocation up uniformly (isolated variant 2, also variant 4), and in proportion to population times marginal utility (variant 3). The temperature row is the change from the proposal's own 2100 warming, which is 2.16°C for Wolfram et al. and 1.80°C for Banerjee, Duflo & Greenstone. World welfare is the NPV of 28 inequality-averse global EDE consumption ( $\eta = 1.5$ ); world consumption is mean consumption per capita, which weights every person's consumption equally.

Table A5: Club members left below their welfare under the proposal, by variant

| Variant                                                             | Wolfram et al. (of 47) |       | Banerjee, Duflo & Greenstone (of 179) |       |
|---------------------------------------------------------------------|------------------------|-------|---------------------------------------|-------|
|                                                                     | isolated               | joint | isolated                              | joint |
| <i>Members losing on equally-distributed-equivalent consumption</i> |                        |       |                                       |       |
| Variant 1                                                           | 10                     | 1     | 41                                    | 36    |
| Variant 2                                                           | 7                      | 2     | 61                                    | 36    |
| Variant 3                                                           | 2                      | 2     | 26                                    | 28    |
| Variant 4                                                           | 7                      | 9     | 61                                    | 64    |
| <i>Members losing on mean consumption per capita</i>                |                        |       |                                       |       |
| Variant 1                                                           | 24                     | 0     | 33                                    | 1     |
| Variant 2                                                           | 4                      | 0     | 28                                    | 0     |
| Variant 3                                                           | 0                      | 2     | 0                                     | 11    |
| Variant 4                                                           | 4                      | 0     | 28                                    | 0     |

Note: a member counts as losing when its NPV of consumption over 2030–2100 falls short of what it obtains under the proposal itself. Variant 1 makes every member indifferent by construction, so any count there is numerical noise. Variants 2 to 4 return the same surplus of rights under different sharing rules: a split chosen to minimise the largest relative shortfall (2), one proportional to population times marginal utility (3), and uniform scaling (4).

## C Welfare or consumption, and the Equal Right schedule

Table 2 reports every allocation twice: solved so that each member keeps its welfare (the net present value of its equally-distributed-equivalent consumption) and solved so that it keeps its total consumption (both population-weighted). The two can move in opposite directions for the same country, for a within-country reason. Carbon costs fall on deciles in proportion to income raised to an elasticity below one, which is regressive, and revenue is returned in proportion to  $c^\eta$ , which is regressive too. More domestic pricing therefore widens within-country inequality. A member held exactly at its total consumption can thus gain or lose on the inequality-averse measure, and conversely. Because this channel depends on the model’s assumptions about the domestic incidence of carbon prices, while the question of this paper is between-country, the main text reads the consumption column as the benchmark and reports the welfare column alongside it.

The Equal Right schedule is designed to rise by about 16% a year, far faster than the 5% we assume for the other two. Table A6 reports it on both paths. On its own path, the schedule pushes most members to full abatement by 2050 and reaches 1.48°C of warming; its club price is far above the other schedules’, and its allocations are compressed towards an equal per capita share, the dividend of the

global fund weighing more heavily the higher the price. The dividend of uniform pricing is larger still: replacing the schedule with the club price and the equivalent allocation cuts the club's emissions by 18.4%, against 15.2% on the 5%-a-year path. Its effect on 2100 warming is smaller ( $-0.013^{\circ}\text{C}$  against  $-0.037^{\circ}\text{C}$ ), because a schedule that has already driven emissions to zero by mid-century leaves little room for a further cut in cumulative terms.

Table A6: The Equal Right schedule on its own escalation path and at 5% a year

| Country                                                                  | Equal Right |                      |                      | Equal Right, 5%/yr |                      |                      |
|--------------------------------------------------------------------------|-------------|----------------------|----------------------|--------------------|----------------------|----------------------|
|                                                                          | $p$         | $\rho^{\text{welf}}$ | $\rho^{\text{cons}}$ | $p$                | $\rho^{\text{welf}}$ | $\rho^{\text{cons}}$ |
| Russia                                                                   | 513         | 0.06                 | 0.22                 | 240                | -0.4                 | -0.04                |
| United States                                                            | 513         | -0.21                | 0.05                 | 240                | -0.96                | -0.35                |
| European Union                                                           | 496         | 0.42                 | 0.45                 | 232                | 0.49                 | 0.52                 |
| China                                                                    | 231         | 1.06                 | 0.98                 | 108                | 1.11                 | 1.03                 |
| Turkey                                                                   | 231         | 0.9                  | 0.84                 | 108                | 1.04                 | 0.95                 |
| Brazil                                                                   | 231         | 0.74                 | 0.68                 | 108                | 0.82                 | 0.74                 |
| Indonesia                                                                | 205         | 0.84                 | 0.79                 | 96                 | 0.95                 | 0.89                 |
| India                                                                    | 90          | 1.17                 | 1.1                  | 42                 | 1.34                 | 1.22                 |
| Nigeria                                                                  | 90          | 0.75                 | 0.72                 | 42                 | 0.79                 | 0.75                 |
| DR Congo                                                                 | 32          | 0.65                 | 0.63                 | 15                 | 0.69                 | 0.66                 |
| <i>Reduced emissions: every member as well off as under the proposal</i> |             |                      |                      |                    |                      |                      |
| World temp. 2100, change ( $^{\circ}\text{C}$ )                          |             | -0.011               | -0.013               |                    | -0.030               | -0.037               |
| Emissions change in the coalition (%)                                    |             | -15.7%               | -18.4%               |                    | -12.4%               | -15.2%               |
| World welfare gain (%)                                                   |             | 0.073%               | 0.068%               |                    | 0.160%               | 0.120%               |
| World consumption gain (%)                                               |             | 0.051%               | 0.045%               |                    | 0.122%               | 0.104%               |
| Members losing on welfare                                                |             | 1                    | 5                    |                    | 4                    | 29                   |
| Members losing on consumption                                            |             | 13                   | 2                    |                    | 23                   | 4                    |
| <i>Increased consumption: coalition emissions as under the proposal</i>  |             |                      |                      |                    |                      |                      |
| World welfare gain (%)                                                   |             | 0.079%               | 0.064%               |                    | 0.148%               | 0.076%               |
| World consumption gain (%)                                               |             | 0.080%               | 0.079%               |                    | 0.175%               | 0.170%               |
| Smallest member gain, welfare                                            |             | 0.057%               | -0.008%              |                    | 0.138%               | -0.111%              |
| Smallest member gain, consumption                                        |             | -0.012%              | 0.079%               |                    | -0.399%              | 0.173%               |
| Members losing on welfare                                                |             | 0                    | 1                    |                    | 0                    | 13                   |
| Members losing on consumption                                            |             | 1                    | 0                    |                    | 8                    | 0                    |

Note: As Table 2, for the Equal Right schedule of price floors grown at its own rate (about 16% a year) and at the 5% a year assumed for the other schedules. Both are run with the proposal's global fund: the revenue of every country is pooled and returned as an equal per capita dividend to the world population.